Data Semantics - Assignment 4 - NER and NEL

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General Information

This assignment is about Named Entity Recognition (NER) and Named Entity Linking (NEL), two building blocks of information extraction solutions centered around entities and supporting knowledge base population tasks.

As a reference for this assignment, consider the **notebook** from the hands-on lesson on Information Extraction. A **dataset** (*datasets_partitions_for_assignment)* is made available for this assignment[1](the zip also contains the notebook). The dataset contains 29 documents of the AIDA benchmark [2] (a benchmark used to train and test NER and NEL algorithms). The documents are in GateNLP format[2] and contain an annotation set named *gold* with the gold-standard annotation you will need for the evaluation.

To complete the assignment, each student is required to choose **only two** documents from the dataset. To load the chosen documents with gatenlp, you can use the following code:

```
import json
from gatenlp import Document

gold_path = "gold_NEL.json" # reassign this variable
    with your dataset

with open(gold_path, "r") as fd:
  gold_dict = json.load(fd)

gold_annotations_doc = Document.from_dict(gold_dict)
gold_annotations_doc # visualize the doc in a notebook

# Explore text and annotations
print("Text:", gold_annotations_doc.text)
print("Annotations:")
print("start", "end", "type", "text", "link")
```

[1] https://drive.google.com/file/d/1BHVQrfJo3Ex9R34EGtPSia2A2v2KD6_I/view?usp=sharing

[2] https://gatenlp.readthedocs.io

```
for ann in gold_annotations_doc.annset("gold"):
    anntext =
        gold_annotations_doc.text[ann.start:ann.end]
    print(ann.start, ann.end, ann.type,
            anntext, ann.features["link"])
```

The assignment consists in completing the following instructions for the two documents of choice.

Task 1: NER

- Load and visualize the chosen documents.
- Apply a spaCy NER pipeline on the text.
- Calculate precision, recall, and F1-score for the NER predictions by comparing with the gold standard spans with exact matching criterion (i.e., a prediction is correct only if start, end, and type are correct).
- Note that spacy types are different from the gold-standard ones, so you need to map them. For instance, you can map GPE (GeoPolitical Entity) to LOC.
- Visualize (e.g. using GateNLP) and analyze qualitatively some of the errors. Try to comment on "can we find different error categories?" and "why are these examples misrecognized/misclassified?" (note that there is no correct answer).

Task 2: NEL

- Load and visualize the chosen documents.
- Perform NER and NEL. You can use dbpedia-spotlight[3] (but it seems the API is not working), or the *Wikipedia-based NEL task implementation* available in the notebook from the hands-on lesson.
- Evaluate the results of NEL.
  - Ignore NIL entities; consider only the entities that exist in the knowledge base (dbpedia);
  - You find the correct entity in the *gold* annotation set by accessing the annotation features (as shown in the example code); the gold standard is using wikipedia while dbpedia spotlight uses dbpedia, but you can use the entity titles for evaluating the predictions.

[3] `https://spacy.io/universe/project/spacy-dbpedia-spotlight`

- Follow the "entity tagging" formulation [1], and calculate *precision*, *recall*, and *F1 score* as described in the following example.
- The entity tagging formulation evaluates whether a NEL system correctly identifies that a document is about certain entities. This approach ignores the number of times each entity is mentioned and the positions of their mentions. Below is an example of evaluating entity linking using this formulation.

Example Text The entity mentions are in bold.

*"**Ronaldo** was born in **Brazil** and started his career at **Cruzeiro Esporte Clube**. Later, **Ronaldo** moved in **Barcelona**."*

This text is about the following entities: Ronaldo (Brazilian footballer), Brazil, Cruzeiro Esporte Clube, and Barcelona. So the expected set of entities (Gold Entities) G_E is:

$$G_E = \{\text{Ronaldo (Brazilian footballer)}, \text{Brazil}, \text{Cruzeiro Esporte Clube}, \text{Barcelona}\};$$

Example Prediction Our system missed *Barcelona* and *Brazil*, and misclassifying the sport club (Cruzeiro Esporte Clube) with the municipality Cruzeiro, São Paulo; the predicted entities P_E are as follows:

$$P_E = \{\text{Ronaldo (Brazilian footballer)}, \text{Cruzeiro Esporte Clube}, \text{Cruzeiro São Paulo}\};$$

In the entity tagging formulation we calculate *precision*, *recall*, and *F1 score* with the following formulas, where the operator $|X|$ is the cardinality of the set X and $A \cap B$ is the intersection of set A and B:

$$P = \frac{|P_E \cap G_E|}{|P_E|} \tag{1}$$

$$R = \frac{|P_E \cap G_E|}{|G_E|} \tag{2}$$

$$F1 = 2 * \frac{P * R}{P + R} \tag{3}$$

Thus, in this example:

$$|P_E \cap G_E| = |\{\text{Ronaldo (Brazilian footballer)}, \text{Cruzeiro Esporte Clube}\}| = 2$$

$$|P_E| = 3$$

$$|G_E| = 4$$

$$P = \frac{2}{3} = 0.67$$
$$R = \frac{2}{4} = 0.5$$
$$F1 = 0.57.$$

References

[1] Christina Du, Kashyap Popat, Louis Martin, and Fabio Petroni. Entity tagging: Extracting entities in text without mention supervision. arXiv:2209.06148, 2022.

[2] Mohamed Amir Yosef, Johannes Hoffart, Ilaria Bordino, Marc Spaniol, and Gerhard Weikum. Aida: an online tool for accurate disambiguation of named entities in text and tables. *Proc. VLDB Endow.*, 2011.